

Assignment – 6

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Python 3.13.3

Github Link(containing code files):

<https://github.com/harshita-0310/AI-ML-Applications-in-Civil-Engineering-CE29208/tree/main/Assignment-06>

Title

ANN-Based One-Month Ahead Streamflow Forecasting for Bhadra River Basin

Introduction

Accurate prediction of streamflow is essential for effective water resources planning, flood management, irrigation scheduling, and reservoir operation. Hydrological systems exhibit strong temporal variability and nonlinear behaviour influenced by precipitation, catchment storage, and routing processes. Traditional statistical models often fail to capture such nonlinear relationships.

Artificial Neural Networks (ANNs) have emerged as powerful data-driven tools capable of modelling nonlinear dependencies between hydrometeorological variables. In this study, an ANN-based forecasting framework is developed to predict one-month-ahead streamflow in the Bhadra river basin using historical streamflow and rainfall data. Lagged inputs of both variables are incorporated to capture hydrological memory effects and rainfall–runoff dynamics.

A sensitivity analysis is conducted by varying lag inputs from one to six months to evaluate the influence of antecedent conditions on prediction accuracy.

Objective

The main objectives of this assignment are:

- To develop an ANN model for one-month-ahead streamflow prediction.

- To incorporate lagged streamflow and rainfall as input variables.
- To analyze the sensitivity of model performance with varying lag periods (1–6 months).
- To evaluate performance using statistical metrics such as RMSE and correlation coefficient.
- To identify the optimal lag structure for accurate forecasting.

Dataset Description

Monthly time-series data of streamflow and precipitation for the Bhadra river basin were used in this study.

The dataset includes:

- Date (monthly resolution)
- Streamflow values (m^3/s)
- Precipitation (mm)

The streamflow series exhibits strong seasonal characteristics due to monsoon rainfall patterns. Rainfall serves as an exogenous input influencing runoff generation, catchment storage, and flow routing.

Data preprocessing steps included:

- Parsing date information
- Converting variables to numeric format
- Removing missing or invalid observations
- Extracting calendar month for seasonal normalization

This ensures data quality and consistency before model training.

Input–Output Structure

To capture hydrological memory, lagged values of both streamflow and precipitation were used as model inputs.

For each time step t :

- Inputs:
 - Streamflow at previous months: $Q(t-1)$, $Q(t-2)$, ..., $Q(t-\text{lag})$
 - Precipitation at previous months: $P(t-1)$, $P(t-2)$, ..., $P(t-\text{lag})$
- Output:
 - Streamflow at current time: $Q(t)$

This multivariate lagged structure allows the ANN to learn delayed rainfall-runoff responses and storage effects within the basin.

Data Pre-processing and Standardization

Streamflow data shows strong seasonality. Hence, month-wise standardization was applied.

For each month m :

$$Q_{\text{std}} = (Q_m - \mu_m) / \sigma_m$$

where:

- μ_m = monthly mean
- σ_m = monthly standard deviation

This standardization ensures:

- all variables lie on comparable scales
- ANN learning is unbiased toward high-magnitude values
- seasonal variability is preserved

Only training data statistics were used to avoid data leakage.

ANN Architecture Used

A Multi-Layer Perceptron (MLP) neural network was employed.

Model characteristics:

- Type: Feedforward ANN (MLPRegressor)
- Hidden layers: Two
- Neurons: (8, 4)
- Activation function: ReLU
- Optimizer: Adam
- Learning rate: 0.001
- Batch size: 32
- Maximum iterations: 8000
- Random state: 42

The two hidden layers enable the model to learn nonlinear rainfall–runoff relationships.

Overall Methodological Flow

The overall modelling framework consisted of the following steps:

Data Pre-processing

- Monthly standardization applied
- Missing values removed
- Lagged variables generated

Model Training

- Data split into training (70%) and testing (30%) sets
- ANN trained using lagged inputs
- Mean Squared Error used as loss function

Model Evaluation

- Correlation coefficient (R) used to measure association
- RMSE used to quantify prediction error
- Higher R and lower RMSE indicate better model performance

Results Visualization

- Observed vs predicted time series plotted
- Scatter plots generated
- Peak and low flows examined

Sensitivity Analysis of Lag Inputs

Lag inputs from 1 to 6 months were tested to understand their impact on prediction accuracy.

The analysis showed:

- Performance improved initially with increasing lag
- Optimal lag observed at three months
- Beyond three months, additional lags introduced noise
- Overfitting observed at higher lag values

This highlights the delayed rainfall-runoff response of the Bhadra basin.

Model Training and Testing Strategy

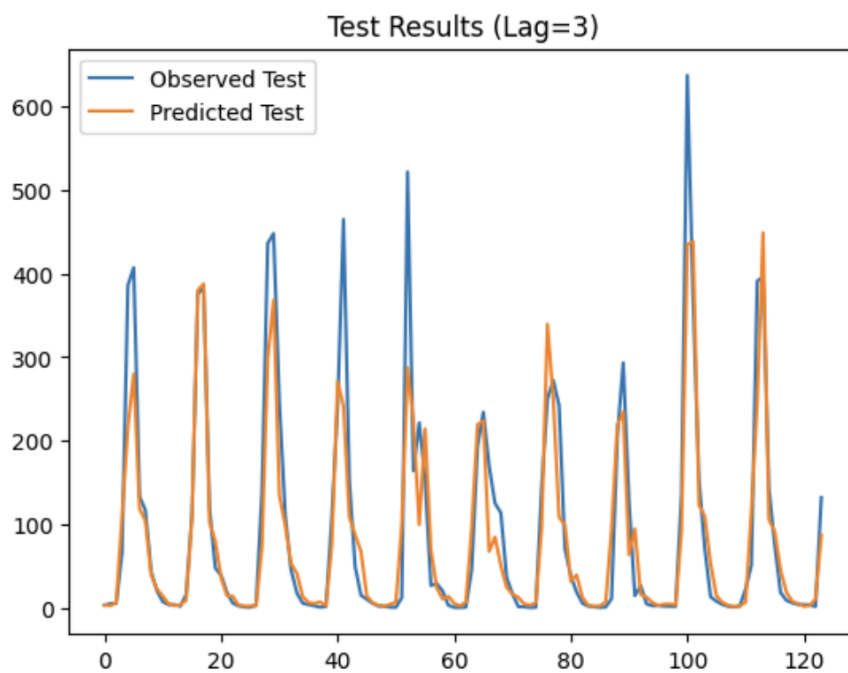
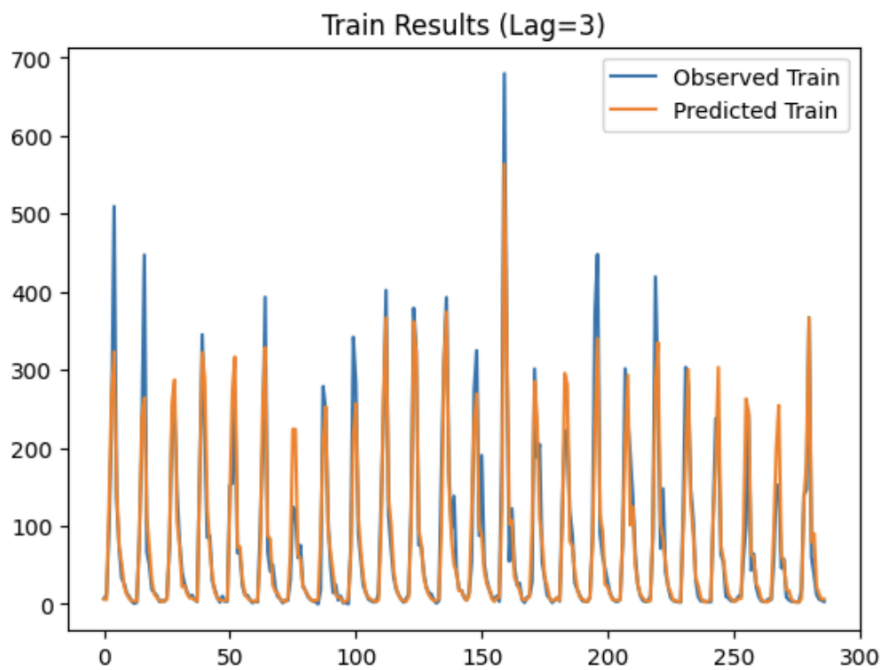
A time-ordered split was used:

- Training dataset: ~70%
- Testing dataset: ~30%

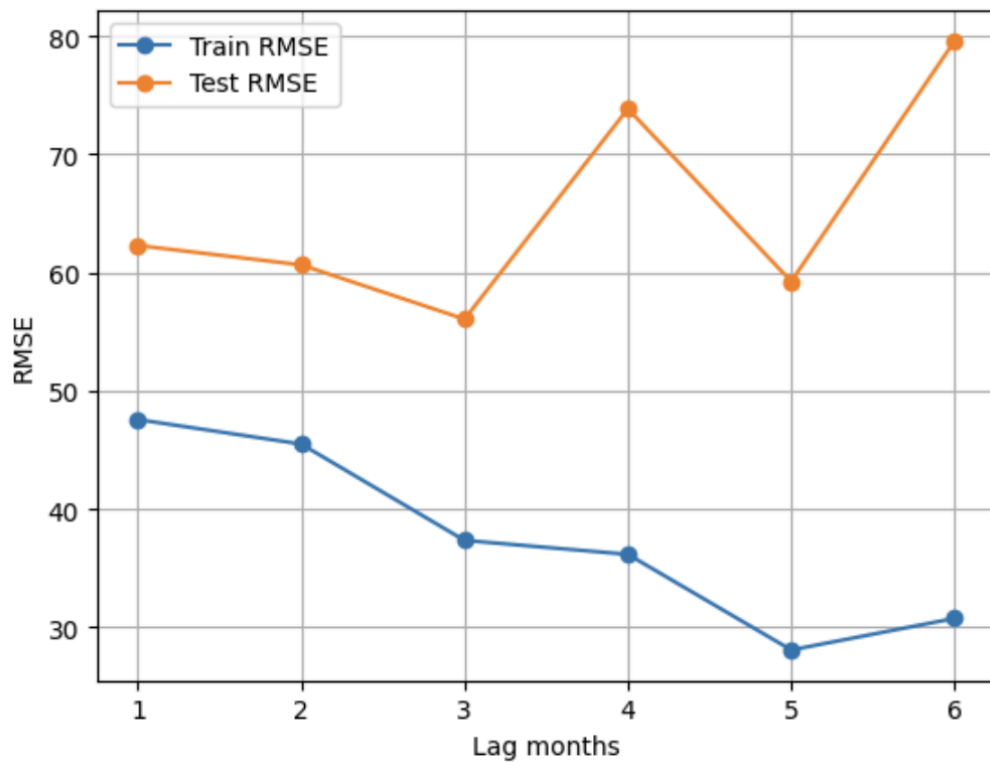
No random shuffling was performed to preserve temporal dependency. Model hyperparameters were tuned only on training data. Testing data was used solely for performance evaluation.

Results visualization through time series and scatter plots

Observed vs Predicted Streamflow (Time Series)



Lag Sensitivity Analysis Using RMSE



Comparison of Training and Testing Performance

These headings perfectly match your two outputs:

- Time series prediction plots
- RMSE vs lag performance plot

They also sound proper for evaluation and viva.

Results and Discussion

The ANN model successfully captured the nonlinear relationship between rainfall and streamflow. The inclusion of lagged precipitation significantly improved forecasting accuracy.

The optimal lag of three months indicates a delayed rainfall–runoff response due to catchment storage and routing mechanisms.

However, peak flows were not captured perfectly, suggesting:

- complex hydrological interactions
- nonlinear rainfall intensity effects
- influence of additional meteorological factors

Model performance degraded at higher lags due to noise and overfitting.

Conclusion

An ANN-based forecasting model was developed to predict one-month-ahead streamflow in the Bhadra river basin. Lagged streamflow and precipitation inputs significantly enhanced model performance.

Sensitivity analysis revealed that a three-month lag provides the best predictive accuracy, highlighting delayed hydrological response. The model demonstrated strong potential for water resource planning and flood management.

ANN models provide a reliable data-driven framework for hydrological forecasting when properly trained and evaluated
